Student Government (SG) · DAUST

Finance Tracker

Technical brief for the IT Committee

Prepared by the Treasurer, Mbaye N. Diouf · Version 1.1 (draft for review) · 12 October 2026

1. Purpose

The SG records its transactions in an Excel file. Entering every transaction there is hard to keep up, so at the end of the term a correct financial review cannot be built. Finance Tracker replaces it with a ledger: every movement of money is one transaction, and every balance is calculated from transactions.

At any moment the app must show (1) how much money the SG holds across its accounts and (2) how much the school still owes the SG. Income from the SG's websites is recorded automatically. Everything else is recorded manually, always with a reason, so that the end-of-term review is a report and not a reconstruction.

In scope for version 1

- Ledger with live balances for Naboopay, Wave, Orange Money, Cash, and the balance held by the university.
- Manual entry of any income, expense, or transfer, from a phone.
- Automatic income from the shop website and DaBus, labeled by the system. Any future SG website can be added the same way.
- Naboopay payouts, fees, and unmatched payments.
- Labels (category, event, source), a review inbox, filters, and CSV export.
- Reconciliation against real account balances, and an end-of-term review report.
- Login for finance committee officers.
- Recurring expense templates that create monthly drafts for confirmation (build after the core ledger, milestone 7).

Out of scope for version 1

- Moving real money, or any bank or mobile-money connection. The app records; it does not pay.
- Generating withdrawal request letters and budget documents (a later phase).
- Access for other SG members or the administration.
- Importing the old Excel history (see section 12).
- Recurring income or subscriptions. No such income source exists yet. If one appears, its payments arrive through the normal intake (section 6).

2. Design rules

These rules decide how the app behaves. If a technical choice conflicts with one of them, the rule wins.

1. **Balances are computed, never typed.** A balance is always the result of the opening balance plus the transactions. If a transaction is missing, the balance is wrong, which is why recording must be fast.
2. **Every transaction has an account, an amount, a reason, a category, and a recorder.** Automatic income gets its reason and category from its source. Manual entries must be given a reason; the form does not save without one.

3. **Nothing is deleted.** A mistake is corrected with a reversal entry linked to the original. Any edit is written to an audit log.
4. **One capture point per payment.** Shop and DaBus income is recorded by the shop or DaBus backend. The Naboopay feed must never create the same income a second time.
5. **Transfers are not income or expenses.** Moving money between the SG's own accounts changes where money is, not how much the SG has.
6. **Amounts are whole FCFA.** Store integers, no decimals. Display with spaces as thousands separator: 253 650 FCFA.
7. **Intake is idempotent.** Sending the same record twice records it once.

3. Accounts

Account	What it is	How money gets in	Balance
Naboopay	Online payment gateway	All shop and DaBus payments (automatic)	Opening balance + transactions
Wave	Mobile money	Payouts from Naboopay; manual entries	Opening balance + transactions
Orange Money (OM)	Mobile money	Payouts from Naboopay; school withdrawals paid to OM; manual entries	Opening balance + transactions
Cash	Cash held by the SG	Payouts converted to cash; cash events; manual entries	Opening balance + transactions
School (held by the university)	Money the school owes the SG	School allocations	Allocations minus withdrawals paid

The home screen shows the balance of each of the first four accounts and their total ("held by the SG"). The School balance is shown separately and is never added into that total. Only Naboopay provides automatic data. Wave, Orange Money, and Cash are recorded manually.

4. Transaction types

Type	Effect	Example
Income	Increases the account	A shop sale, a DaBus booking, a sponsor payment
Expense	Decreases the account	Supplies, catering, venue. Provider fees are expenses with the category "Provider fees".
Transfer	Decreases one account and increases another. Total held does not change.	Naboopay payout to Wave
School allocation	Increases the School balance	The university credits the SG its yearly allocation
School withdrawal	When paid: decreases the School balance and increases the destination account	A withdrawal paid out to Orange Money
Adjustment	Increases or decreases an account to match reality. A reason is required.	The difference found during reconciliation

School withdrawals have two states

A withdrawal that the university has approved but not yet paid is **committed**. It is shown on the home screen under the School balance, and it does not change any balance. Only when it is marked **paid** does the School balance go down and the destination account go up, in one operation.

Reversals

A reversal is a normal transaction with the opposite effect and a link to the transaction it cancels (reversalOf). Because of this, the balance formula never needs a special case for reversals.

5. Data model

Tables: **accounts**, **transactions**, **categories**, **events**, **reconciliations**, **users** (finance officers), **auditLog**, **recurringTemplates**. The transactions table is the core.

transactions

Field	Type	Notes
type	enum	income, expense, transfer, allocation, school_withdrawal, adjustment
amount	integer	FCFA, always positive. The direction comes from the type.
occurredAt	timestamp	When the money moved. Defaults to now for manual entries; for automatic entries, the payment time.
accountId	reference	The account affected. For a transfer, the source account.
toAccountId	reference	Transfers and school withdrawals only: the destination account.
reason	text	Required. Shown in every list and report.
categoryId	reference	Required. Items captured without one are given "Unlabeled" and status needs_review.
eventId	reference	Optional (Integration Week, Gala, DAUST Impact, and so on).
source	text	shop, dabus, naboopay, manual, or the key of a future website.
externalId	text	The ID in the source system (for example the order ID). Unique together with source.
status	enum	confirmed, needs_review, committed (approved school withdrawal not yet paid), or draft (generated from a recurring template, awaiting confirmation).
receiptUrl	text	Optional. A photo or PDF of the receipt.
recordedBy	reference or text	The officer who entered it, or the source name for automatic entries.
reversalOf	reference	Optional. The transaction this entry cancels.
recurringId	reference	Optional. The recurring template that generated this draft. Unique together with the due month.
createdAt	timestamp	When the record was created in the app.

Balance formula

Balance of an account = opening balance + income + incoming transfers + paid school withdrawals into it + positive adjustments − expenses − outgoing transfers − negative adjustments. School balance = allocations − paid school withdrawals. Committed withdrawals and drafts are displayed but never included.

6. Automatic income: shop and DaBus

Customers pay through Naboopay. Both backends are controlled by the SG, so each of them reports every confirmed payment to the tracker.

Flow

1. The customer pays through Naboopay.
2. Naboopay confirms the payment to the shop or DaBus backend.
3. Only then, the backend sends one income record to the tracker. An unpaid or pending order is never sent.
4. The tracker saves it as income on the Naboopay account, labeled with the source's default category, with status confirmed.

Endpoint

An HTTPS POST endpoint, for example /intake/income. Each source has its own secret key sent in a header, so one source can be revoked without affecting the others. Keys are stored as environment secrets and never in the front end.

Field	Required	Description
source	Yes	Identified by the key: shop, dabus, or a future site.
externalId	Yes	Order or booking ID in the source system.
amount	Yes	Whole FCFA, positive.
paidAt	Yes	Time Naboopay confirmed the payment.
description	Yes	What was sold or booked. Becomes the reason.
naboopayReference	Yes	The Naboopay payment reference, used to match the Naboopay feed (section 7).
category	No	Overrides the source's default category.
event	No	Event name or ID, if the sale belongs to an event.

Required behavior

- **Duplicates:** if (source, externalId) already exists, return the existing record and create nothing.
- **Validation:** reject non-integer or non-positive amounts, and unknown keys (401).
- **Retries:** if the tracker is unreachable when a payment is confirmed, the backend must queue the record and retry. A sale must not be lost. Retrying is safe because of the duplicate rule.
- **Refunds:** the source calls a refund endpoint with the original externalId. The tracker creates a reversal of the original income.
- **New websites:** a new site needs only a new key and the same payload. No tracker code change.

7. Naboopay feed

Naboopay is the only provider that gives automatic data. The feed does not create income, because shop and DaBus already did (rule 4). It has three jobs:

Job	What the tracker does
Payouts	A payout from Naboopay to Wave, OM, or Cash is recorded as a transfer.
Fees	Naboopay fees are recorded as expenses (category "Provider fees") so the computed balance matches the real one.

Job	What the tracker does
Safety net	A Naboopay payment that matches no income record by naboopayReference is saved with status needs_review and shown in the review inbox.

If the feed does not expose payouts or fees, those are entered manually and the rest still works. Build this after the manual flow and the shop/DaBus intake (section 14), because it depends on the open questions in section 15.

8. Manual entry

Everything not captured automatically is entered by hand: expenses, cash income, transfers, school allocations and withdrawals, adjustments.

Required	Optional
Type, amount, account, reason, category	Event, receipt photo, date (defaults to now)

- **Phone first.** One-handed use, large inputs, usable in under 10 seconds.
- **Faster repeat entry.** Suggest recent reasons and categories as the officer types.
- **Offline queueing (should have).** Entries made without a connection are stored on the device and synced when it returns. The IT Committee should confirm feasibility for the first release.

Recurring expenses (version 1.1)

Regular payments such as hosting, a domain, a software plan, or a phone or Wi-Fi bill are set up once as a recurring template: reason, amount, account, category, optional event, and day of the month. Templates only prepare entries. They never post an expense by themselves.

- **Monthly draft.** On the due day the tracker creates one draft expense from each active template, dated the due date. At most one draft exists per template per month.
- **A draft changes no balance.** It counts only after an officer confirms it. If a payment failed or the amount changed, the officer corrects the amount or date when confirming, or skips the month. A skipped month creates no transaction.
- **Visibility.** Open drafts appear in the Needs review inbox and in the badge on the home screen. Drafts past their due date are highlighted, which also serves as a payment reminder.
- **Editing a template** changes future drafts only. Existing drafts and confirmed transactions are untouched. A template can be deactivated, not deleted.

Recurring income is not covered: the SG has no subscription income source at the moment (see section 1).

9. Screens (version 1)

Screen	Purpose and content
Sign in	Finance committee officers only.
Home	Balance of Naboopay, Wave, OM, Cash and their total. School balance with the committed amount under it. Latest transactions. A badge with the number of items needing review. Last reconciliation date per account.
Add transaction	The manual entry form (section 8).
Needs review	Everything captured without a label, unmatched Naboopay payments, and recurring-expense drafts awaiting confirmation. The aim is to keep it at zero.

Screen	Purpose and content
Recurring	List of recurring templates (version 1.1): create, edit, deactivate. Shows the next due date for each.
Transactions	Full list. Filters: date, account, category, event, source, type. CSV export.
Event view	Income, expenses, and net for one event, grouped by category.
Reconcile	See section 10.
Review report	Summary for a chosen period: totals by category and by event, balances at the start and end, the School account movement, and the reconciliation history. This is the report that replaces the end-of-term reconstruction.

10. Reconciliation

1. The officer opens Reconcile and picks an account.
2. They enter the real balance: the Naboopay dashboard, the Wave or OM app, or a physical count of the cash.
3. The app shows its computed balance and the difference, and stores the check in the reconciliations table.
4. If there is a difference, the officer finds the missing transaction or creates an adjustment with a written reason.

Reconciliation should be done weekly and after every event. The home screen shows the date of the last check per account, so a missed one is visible.

11. Users and access

- Finance committee officers only, each with a personal login.
- Every transaction stores who recorded it. Every edit is written to the audit log with who, when, and what changed.
- All officers have the same rights in version 1.
- Intake keys (section 6) are separate from user logins.
- Later: President approval of withdrawals, and read-only access for other members.

12. Starting the app (cutover)

The existing Excel file is too inconsistent to import. The app starts from verified opening balances, and the Excel stays as an archive.

1. Choose a cutover date, ideally the day the app goes live.
2. Set the opening balance of each account from reality: the Naboopay, Wave, and OM apps as they stand, and a physical count for Cash.
3. Set the School balance from a statement from the university showing allocations credited and withdrawals paid.
4. From that date, every transaction goes through the app. The first reconciliations confirm the starting point.
5. Optional: one summary entry per past event (total income, total expenses) from the Excel figures that are trusted, so event totals exist for the review.

13. Suggested stack and branding

- **Suggested stack:** Convex (database, authentication, HTTP endpoints, live queries) with a React web app that works well on phones. Live queries mean balances update on every device as soon as a transaction is saved. The IT Committee may choose differently, provided the rules in section 2 hold.
- **Hosting and code:** in the IT Committee's GitHub organization.
- **Look:** IBM Plex Sans, FCFA amounts, and DAUST Student Government branding, consistent with the other SG tools.

Proposed starting categories (to be agreed with the Treasurer)

Direction	Categories
Income	Shop sales, DaBus, Sponsorship, Fundraising, Other income
Expense	Supplies, Catering, Venue, Decorations, Merchandise purchase, Transport, Provider fees, Other expense

14. Build order and acceptance checks

#	Milestone	Done when
1	Schema, officer login, manual entry form, live balances	A transaction entered on a phone changes the correct balance on every device immediately. A transfer does not change the total held. A reversal restores the balance.
2	Shop and DaBus intake	The same order sent twice produces one transaction. An unpaid order never appears. A wrong key is rejected. A refund creates a reversal.
3	Needs-review inbox, filters, event view, CSV export	Every captured item without a label is in the inbox. Event totals match the transactions list.
4	Reconciliation and audit log	A difference between real and computed balance is recorded and can be fixed with an adjustment. Every edit appears in the audit log.
5	Naboopay feed	Payouts appear as transfers, fees as expenses, and an unmatched payment appears in the inbox. Not started until section 15 is answered.
6	Review report	The report for a period reproduces the totals in the transactions list.
7	Recurring expenses (version 1.1)	A template creates exactly one draft per month. A draft changes no balance until confirmed. A skipped month creates no transaction. An overdue draft is visible on the home screen.

15. Open items to verify

These must be answered, from the Naboopay documentation or dashboard, before milestone 5. Nothing here has been confirmed yet.

1. How does Naboopay deliver data: webhook, API polling, or export only?
2. Does the payment confirmation carry our order or booking reference, so it can be matched to the shop or DaBus record?
3. Is the amount credited to the balance gross or net of fees?
4. Do payouts and fees appear in the data we can receive, and with what details (amount, destination, time)?

5. How does the shop or DaBus backend receive Naboopay's confirmation today, and where in that code should the call to the tracker go?
6. Which categories and events should the shop and DaBus attach by default? (Treasurer to confirm.)

Contact: Mbaye N. Diouf, Treasurer, DAUST Student Government.